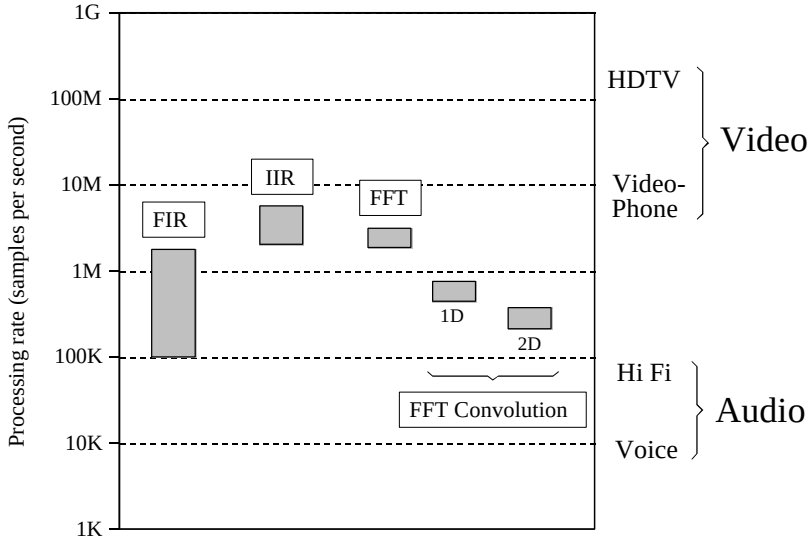

FIGURE 28-11
The speed of DSPs. The throughput of a particular DSP algorithm can be found by dividing the clock rate by the required number of clock cycles per sample. This illustration shows the range of throughput for four common algorithms, executed on a SHARC DSP at a clock speed of 40 MHz.

in mind that microprocessor speed is doubling about every three years. This means you should pay special attention to the *method* we use in this example. The actual numbers are always changing, and you will need to repeat the calculations every time you start a new project. In the world of twenty-first century technology, blink and you are out-of-date!

When it comes to understanding execution time, the SHARC family is one of the easiest DSP to work with. This is because it can carry out a multiply-accumulate operation in a single clock cycle. Since most FIR filters use 25 to 400 coefficients, 25 to 400 clock cycles are required, respectively, for each sample being processed. As previously described, there is a small amount of overhead needed to achieve this loop efficiency (priming the first loop and completing the last loop), but it is negligible when the number of loops is this large. To obtain the throughput of the filter, we can divide the SHARC clock rate (40 MHz at present) by the number of clock cycles required per sample. This gives us a maximum FIR data rate of about 100k to 1.6M samples/second. The calculations can't get much simpler than this! These FIR throughput values are shown in Fig. 28-11.

The calculations are just as easy for recursive filters. Typical IIR filters use about 5 to 17 coefficients. Since these loops are relatively short, we will add a small amount of overhead, say 3 cycles per sample. This results in 8 to 20 clock cycles being required per sample of processed data. For the